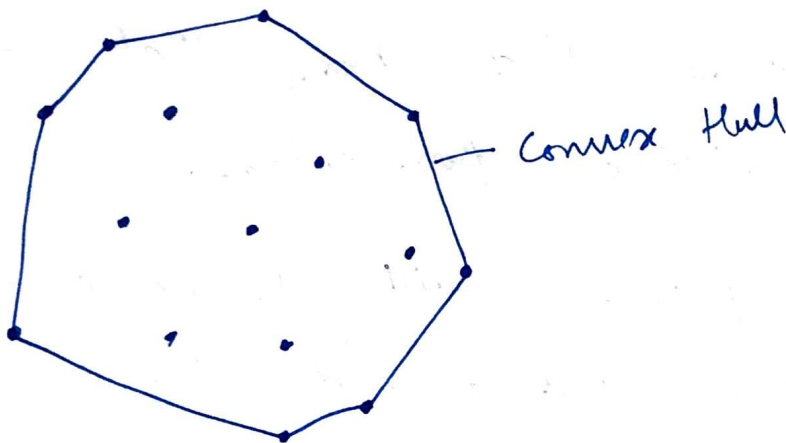


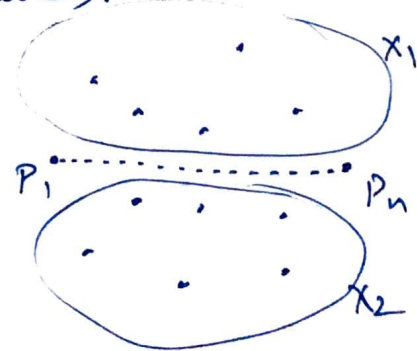
Convex Hull using Divide & Conquer

Given a finite set of points in a 2-D plane, the convex hull is the smallest convex polygon that encloses all the given points such that every line segment joining any two points inside the polygon lies completely inside the polygon.



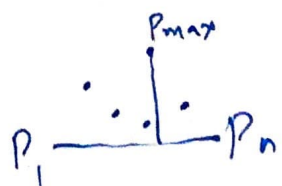
Procedure:

1. Divide the plane X into two parts X_1 (upper) & X_2 (lower) using line $P_1 P_n$ (P_1 is min x -coordinate, & P_n is max x -coordinate). & draw a st. line joining P_1 & P_n .



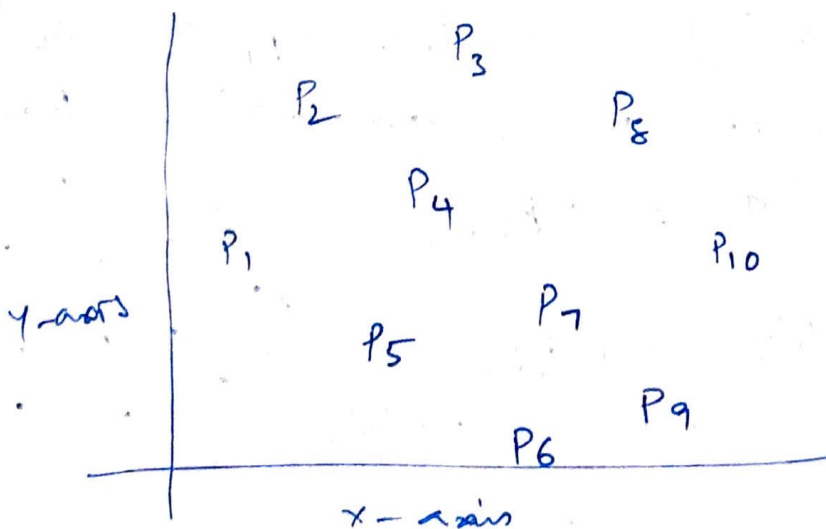
2. If X_1 is empty then $P_1 P_n$ itself forms the upper convex hull.

3. If X_1 is not empty, find the pt. P_{max} that is its perpendicular dist. from line $P_1 P_n$ is max.

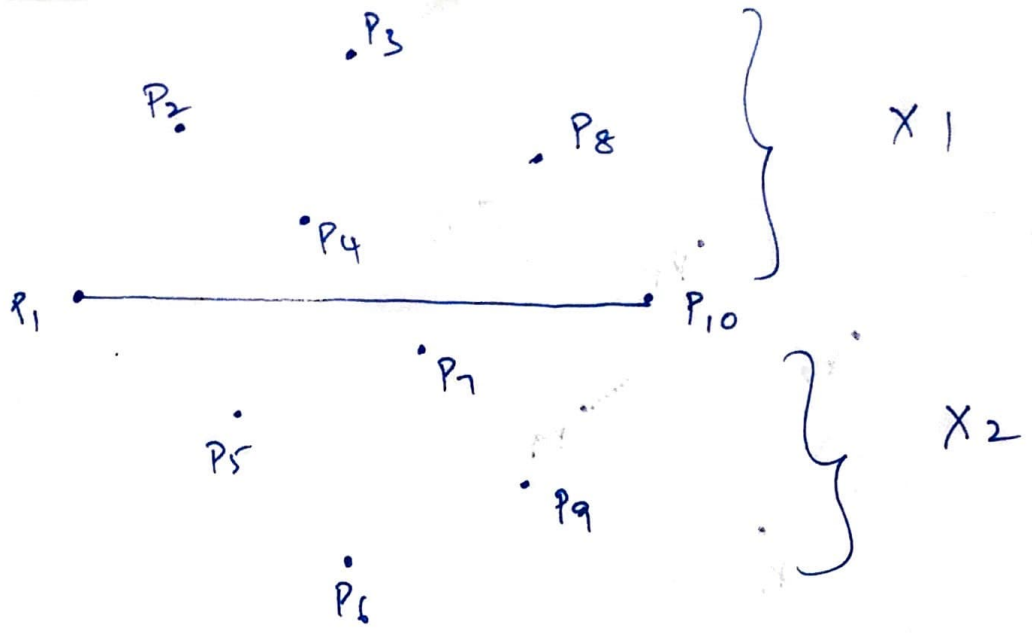


3. If there are more than one point with same distance, select the point that forms max. angle with line $P_1 P_n$.
4. Join $P_1 P_{\max}$ and $P_{\max} P_n$ forming Δ .
5. Now consider $P_1 P_{\max}$ as base line and points left side follow same process. Among all points left of $P_1 P_{\max}$, find $P_{\max}$, that is having max $\perp$ distance from $P_1 P_{\max}$.
6. Repeat the process until all points in upper hull are covered and lies within convex hull.

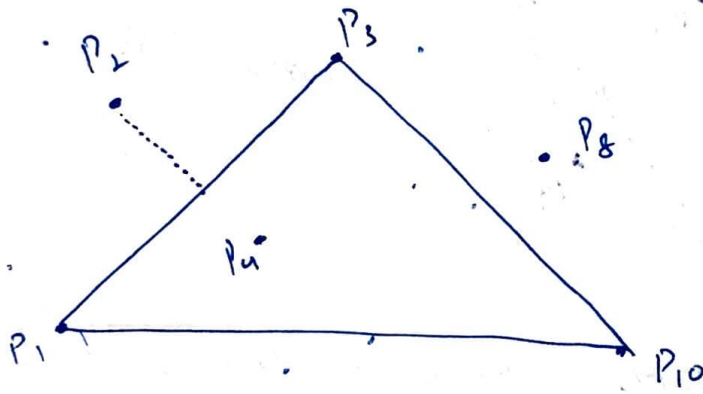
Ex:



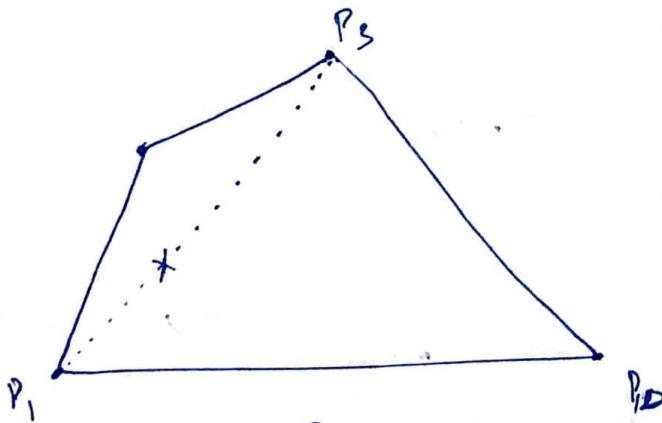
(i)



(ii)

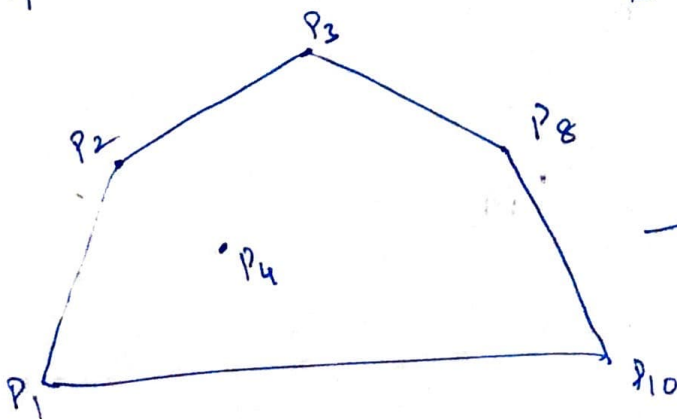


(iii)



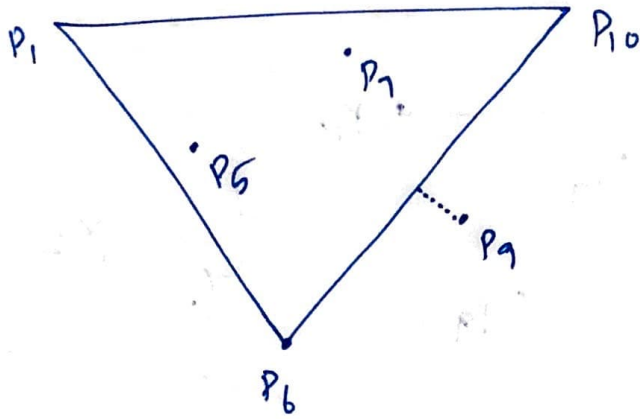
→ upper hull

(iv)

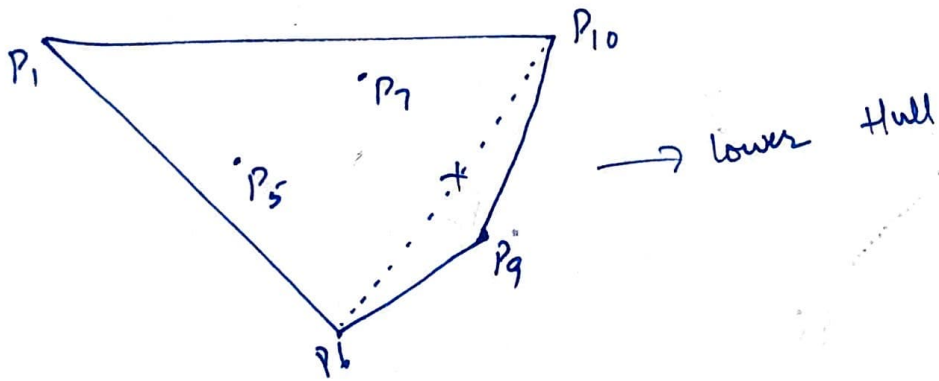


→ upper hull containing all points.

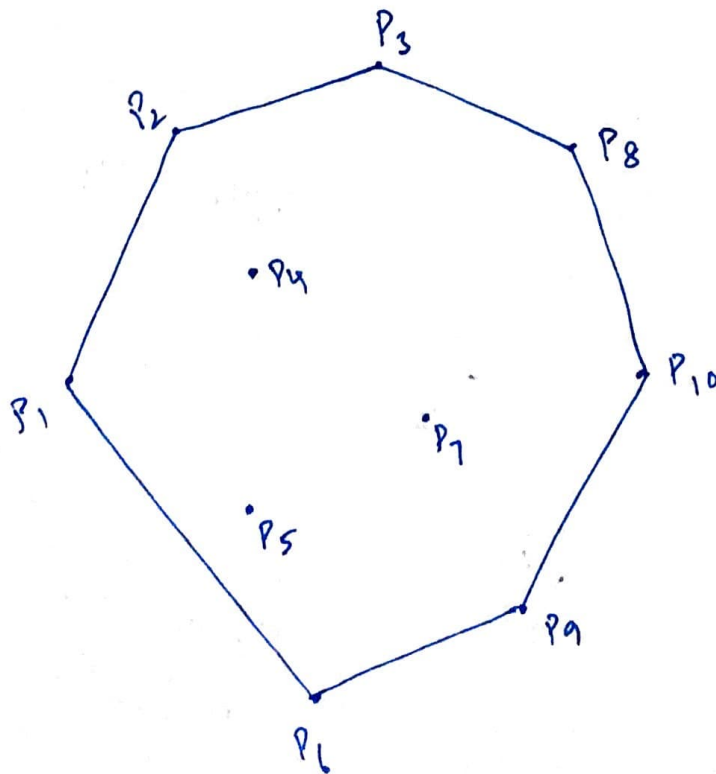
(v)



(vi)



Now merge both upper & lower hull.



→ Convex Hull

Coin Exchange Problem

→ Calculate no. of possible way from given set of coins to get desired sum

Ex: coins = {2, 3, 5} ; Total sum = 7.

Coin \ Sum	0	1	2	3	4	5	6	7
2	1	0	1	0	1	0	1	0
3	1	0	1	1	1	1	2	1
5	1	0	1	1	1	2	2	2

if coins > sum then copy from above row,

else

- exclude above cell

- include same row (sum - coins)

- Add both.

* Assuming infinite no. of coins *

{ 2, 2, 3 } { 5, 2 }

→ finding min no. of coins.

	0	1	2	3	4	5	6	7
2	0	∞	1	∞	2	∞	3	∞
3	0	∞	1	1	2	2	2	3
5	0	∞	1	1	2	1	2	(2)

if $\text{coins} > \text{sum}$, copy above row values.

else

$\min(\text{above value}, \text{include coins} + \text{value same row (sum-coins)})$

Ex: for sum 3 by coins 3,

$$\min(\infty, 1+0) = 1$$

for sum 4 by coins 3

$$\min(2, 1+\infty) = 2$$

for sum 5 by coins 3

$$\min(\infty, 1+1) = 2$$

$$\min(3, 1+1) = 2$$

$$\min(\infty, 1+2) = 3$$

for sum 5 by coins 5

$$\min(2, 1+0) = 1$$

$$\min(2, 1+\infty) = 2$$

$$\min(3, 1+1) = 2$$

∴ min coins are 2

(2, 5)

Greedy method.

$$\text{Coins} = \{4, 10, 25, \cancel{1, 5, 6, 9}\}$$

$$\text{sum} = 41$$

$(25, 10, 4) \rightarrow$ sol
doesn't
exist.

$$41 - 25 = 16$$

$$16 - 10 = 6$$

$$6 - 4 = 2$$

but actually there is using dynamic programming

$$\star (25, 4, 4, 4, 4) \checkmark$$

$$\text{coins} = \{1, 5, 6, 9\}$$

$$\text{sum} = 11$$

$$11 - 9 = 2$$

$$(9, 1, 1)$$

$$2 - 1 = 1$$

$$1 - 1 = 0$$

3 coins but
not optimal
using greedy

but by dynamic program.

$$\underline{\underline{(5, 6)}} \rightarrow \text{optimal.}$$

Huffman coding

- It is a compression technique used for reduce the size of message
- Data can be compressed and sent over a network to reduce the cost of transmission

Message: B C C A B B D D A E C C B B A E D D C C

(1) Normal message sent without Encoding:

- Size of message is measured in terms of bits

length - 20 characters (it has to be sent using ASCII codes)

ASCII → 8 bits

A	65	01000001
B	66	01000010
C	67	⋮
D	68	⋮
E	69	⋮

if normally the message is sent, each ^{char} bit has 8-bits so $20 \times 8 = \underline{160 \text{ bits}}$.

(ii) Fixed size encoding

- Use our own codes instead of ASCII codes

* 8 bit ASCII codes are for 128 char's.
* But we have only 5 char
So 5 needs only 3 bits.

char	count	code
A	3	000
B	5	001
C	6	010
D	4	011
E	2	100
20		

There are 20 char. in message

$$\text{So } 20 \times 3 = 60 \text{ bits.}$$

$$\frac{5 \times 8 \text{ bit}}{\text{char's}}$$

$$\frac{5 \times 3}{\text{codes}}$$

$$40 + 15 = 55$$

Message - 60 bits

Table - 55 bits

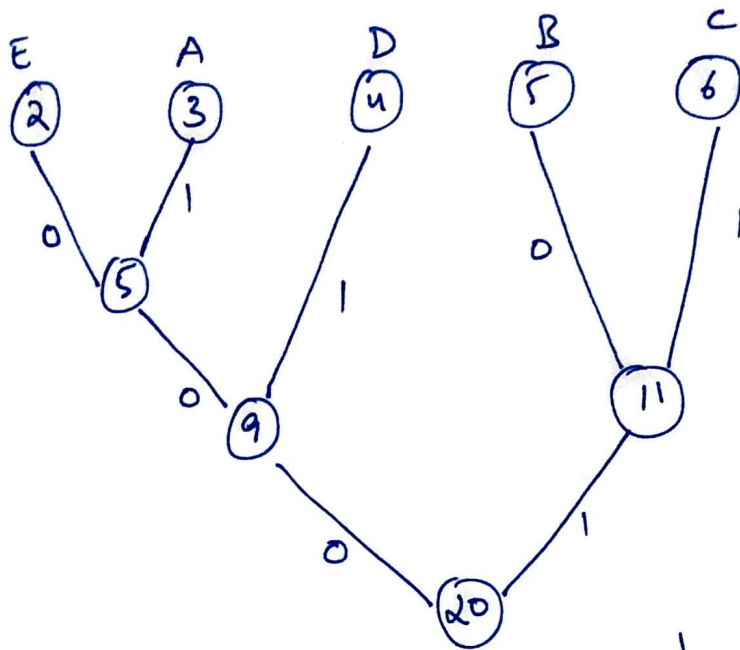
$$\underline{115 \text{ bits.}}$$

(iii) Variable size coding (Huffman man)

→ Arrange char in increasing order of their count.

E	A	D	B	C
2	3	4	5	6

→ merge 2 smallest one and make it one



A $\rightarrow$ 001
 B $\rightarrow$ 00
 C $\rightarrow$ 11
 D $\rightarrow$ 01
 E $\rightarrow$ 000

char	count	code
A	3	001 $3 \times 3 = 9$
B	5	10 $2 \times 5 = 10$
C	6	11 $2 \times 6 = 12$
D	4	01 $2 \times 4 = 8$
E	2	000 $2 \times 3 = 6$
		<u>45</u>

Msg. size - 45 bits.

578 bits
 ↓
 7 chars
 + 12 bits code
 46 + 12 = 52 bits.

$\therefore$ 45 (Msg.)
 52 (codes)
97 bits.